

The Critical Shrinking-Annulus Threshold at the Minimal Tail Clock

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Abstract

The minimal tail-absorbed clock is shorter than the slope-four clock, but its noisy tail is only logarithmically small on the target circle. We determine how rapidly an auxiliary radius may separate from that circle. If $\rho_\sigma = Re^{\eta_\sigma}$ and $h_\sigma = \lceil a_* \log(1/\sigma) \rceil$, the mass-and-cap noisy-tail scale is $e^{a_* L_\sigma \eta_\sigma} / L_\sigma$. For $\eta_\sigma = c \log L_\sigma / L_\sigma$, the sharp information-class threshold is $c_* = 1/a_* = \log(10/7)$. A diagonal multiset consisting of repeated q -atoms saturates all three regimes. The result controls the tail only; the moving head remains open.

1 Minimal clock and moving radius

Put

$$q = \frac{1}{2}, \quad R = \frac{7}{5}, \quad a_* = \frac{1}{\log(1/(qR))} = \frac{1}{\log(10/7)}, \quad L_\sigma = \log(1/\sigma), \quad h_\sigma = \lceil a_* L_\sigma \rceil.$$

Let $\rho_\sigma = Re^{\eta_\sigma}$, where $\eta_\sigma \downarrow 0$, and assume $\rho_\sigma < \rho_*$. For complement traces satisfying $|\tau_{\sigma,n}| \leq 4\sigma^{-1}q^n$, define their tail logarithm beyond h_σ .

2 Threshold theorem

Theorem 2.1. *For $X = H^\infty(\rho_\sigma)$ or $H^2(\rho_\sigma)$, the noisy tail obeys*

$$\|S_\sigma^{\geq h_\sigma}\|_X \leq C \frac{\exp(a_* L_\sigma \eta_\sigma)}{L_\sigma}$$

for all sufficiently small σ , with C independent of σ . The deterministic target tail is smaller:

$$\|T^{\geq h_\sigma}\|_X \leq C' \frac{\sigma^{1-a_* \log(q_*/q)} \exp(a_* L_\sigma \eta_\sigma)}{L_\sigma}.$$

If

$$\eta_\sigma = c \frac{\log L_\sigma}{L_\sigma},$$

then the noisy scale is $L_\sigma^{a_* c - 1}$. Thus it vanishes for $c < c_* = 1/a_*$, is of constant order for $c = c_*$, and grows for $c > c_*$. Here

$$c_* = \log(10/7) = 0.3566749439387324 \dots$$

These three orders are sharp in the mass-and-cap information class.

Proof. Put $x_\sigma = q\rho_\sigma = qRe^{\eta_\sigma}$. The geometric bounds are

$$\|S_\sigma^{\geq h}\|_{H^\infty} \leq \frac{4\sigma^{-1}x_\sigma^h}{h(1-x_\sigma)}, \quad \|S_\sigma^{\geq h}\|_{H^2} \leq \frac{4\sigma^{-1}x_\sigma^h}{h\sqrt{1-x_\sigma^2}}.$$

Since $(qR)^{a_*L_\sigma} = \sigma$, the first estimate follows; the ceiling changes only a bounded factor. The target proof is identical with $48(q_*\rho_\sigma)^h$ and

$$(q_*R)^{a_*L_\sigma} = \sigma^{1-a_*\log(q_*/q)}.$$

The remaining exponent is $0.0531384836\dots > 0$.

For sharpness, take $N_\sigma = \lfloor \sigma^{-1}/q^2 \rfloor$ copies of the atom q on a diagonal operator. Its squared mass is at most σ^{-1} and its trace powers are $N_\sigma q^n$, asymptotic to the upper envelope. The first tail coefficient gives the same lower order in both norms, while the geometric estimates give the matching upper order. $\square$

3 Protocol and boundary

The computation evaluates $c = c_*/2, c_*, 3c_*/2$ at $\sigma = 10^{-80}$, records η_σ, ρ_σ , and the normalized tail scale, and verifies that every reported radius remains below ρ_* . At the critical row the normalized model scale equals one by definition; this means $\Theta(1)$, not convergence of the actual tail to one.

Remark 3.1. The repeated- q diagonal is a saturation model for the available mass-and-cap information, not the measured noisy spectrum. The theorem does not control the moving head and does not prove actual annular nonconvergence. Gates A–E remain false/open.